THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 4: **Expressions using Letter-Numbers** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a = 5$, the value of $a^2 + 3a$ is: 1

(a) 40 **(b)** 39 **(c)** 41 **(d)** 42

2. Value of $7x + 2y - 4$ for $x = 3$, $y = 8$ is: 1

(a) 35 **(b)** 34 **(c)** 33 **(d)** 32

3. Simplify $15x + 2y - 9x + 8y$: 1

(a) $6x + -6y$ **(b)** $16xy$ **(c)** $6x + 10y$ **(d)** $24x + 10y$

4. If $x = 10$, then $2(x - 4) + 9$ equals: 1

(a) 21 **(b)** 22 **(c)** 20 **(d)** 23

5. Subtract $6x + 3$ from $11x + 12$: 1

(a) $5x + 9$ **(b)** $5x + 15$ **(c)** $17x + 15$ **(d)** $-5x + -9$

6. The nth term of a pattern is $3n + 9$. Its 15th term is: 1

(a) 55 **(b)** 56 **(c)** 54 **(d)** 53

7. If a pencil costs ₹x and an eraser ₹y, the cost of 8 pencils and 3 erasers is: 1

(a) $11xy$ **(b)** $8x + 3y$ **(c)** $24(x + y)$ **(d)** $8y + 3x$

8. If $a = 6$, the value of $a^2 + 4a$ is: 1

(a) 59 **(b)** 61 **(c)** 60 **(d)** 62

9. If $x = 5$, the value of $\frac{4x+2}{2}$ is: 1

(a) 11 **(b)** 10 **(c)** 13 **(d)** 12

10. If $x = 4$, the value of $\frac{6x+3}{3}$ is: 1

(a) 8 **(b)** 10 **(c)** 9 **(d)** 11

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the value of $x^2 + 5x - 7$ when $x = 4$. 2

12. Simplify: $5(x + 2) + 8x$. 2

13. Subtract $5x - 4$ from $12x + 9$. 2

14. Find the value of $x^2 + 4x - 3$ when $x = 7$. 2

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SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A number is $12n + 7$. Another is $5n - 6$. Find their sum and difference (first minus second), and evaluate both for $n = 3$. 3

16. Matchsticks make a row of triangles: n triangles need $2n + 1$ sticks. How many sticks for 20 triangles? How many triangles can be made with 61 sticks? 3

17. A cab charges ₹40 plus ₹16 per km. Write the fare for k km. A rider pays ₹152. Find k, and the fare for 12 km. 3

OR

A number is $11n + 2$. Another is $4n - 1$. Find their sum and difference (first minus second), and evaluate both for $n = 6$. 3

18. Matchsticks make a row of triangles: n triangles need $2n + 1$ sticks. How many sticks for 11 triangles? How many triangles can be made with 33 sticks? 3

OR

A cab charges ₹55 plus ₹12 per km. Write the fare for k km. A rider pays ₹163. Find k, and the fare for 14 km. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A shopkeeper sells x kg of rice at ₹38 per kg and y kg of wheat at ₹28 per kg. Write his earning. Find it for $x = 16$, $y = 24$ and the average price per kg (to the nearest rupee). 5

OR

Raju is r years old. His mother is 3 years older than 3 times his age. In 4 years, find the expressions for each age and the sum of ages. Evaluate at $r = 14$. 5

20. A shopkeeper sells x kg of rice at ₹55 per kg and y kg of wheat at ₹35 per kg. Write his earning. Find it for $x = 8$, $y = 12$ and the average price per kg (to the nearest rupee). 5

OR

Raju is r years old. His mother is 7 years older than 3 times his age. In 2 years, find the expressions for each age and the sum of ages. Evaluate at $r = 8$. 5

END OF QUESTION PAPER — Total: 40 Marks

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